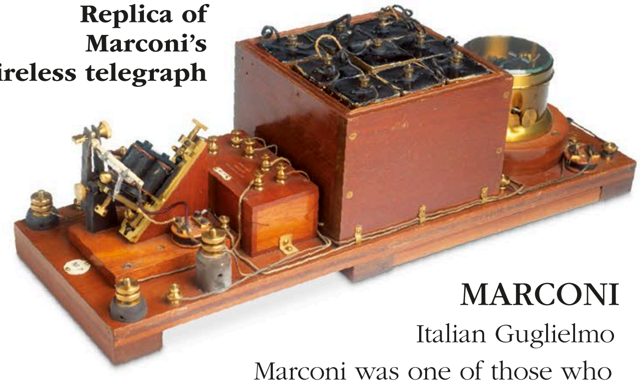
Radio

German physicist Heinrich Hertz discovered radio waves in 1888. He understood they were a form of energy, like light, that traveled in waves, but he did not see any potential practical uses. However, other scientists seized on his discovery and, within a decade, it was possible to send signals around the world without the need for wires.

Early crystal radios required earphones, so only one person could listen at a time.



Replica of Marconi's wireless telegraph

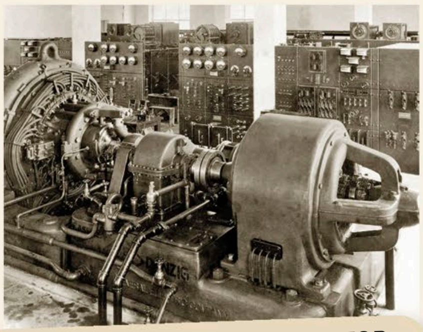
MARCONI

Italian Guglielmo Marconi was one of those who was inspired by Hertz's radio waves discovery. Starting in 1894, his experiments saw him transmit a radio signal a distance of 2 miles (3.2km). In 1896, he received the world's first patent for a radio.

Audion valve with mounting, 1906

THE AUDION

Building on the work of British scientist John Fleming, American inventor Lee de Forest developed the "audion" valve in 1906. This was able to amplify weak electrical signals and was crucial for the development of radio. In the US, de Forest is known as "the father of radio."



ALEXANDERSON ALTERNATOR

The Alexanderson Alternator was a rotating machine invented by electrical engineers Ernst Alexanderson and Reginald Fessenden in 1904. It produced a high-frequency alternating current and was one of the first devices capable of generating continuous radio waves.

Crystal sets produce weak sound, so had to be listened to using earphones.